The $\mathcal{C}_3$ Stratum in Non-DP Universality

Slotting Manna and Conserved-DP, and what currently resists slotting

S. Amarteifio

Percolation Labs

April 2026

Abstract

The non-directed-percolation (non-DP) universality classes catalogued by Ódor (2004) — Manna, voter, conserved-DP (C-DP), and a handful of others — have resisted a unified organising principle. We apply the CFAC stratification theorem (Amarteifio, 2026a), which organises polynomial Dyson–Schwinger equations by the Puiseux order k of their dominant branch point via $\tau = 1 + 1/k + \gamma_k(d)$, where the first two terms are a d-independent skeleton and $\gamma_k(d)$ is a d-dimensional dressing computable from the rank-k bridge function. Two observations: **(i)** The Manna class ($\tau \approx 1.29$) and conserved-DP ($\tau \approx 1.35$) both sit within 3% of the $\mathcal{C}_3$ skeleton $\tau = 4/3$, flanking it from opposite sides — a non-trivial prediction that the two classes are a common $k = 3$ stratum split only by their $\gamma_3(d)$ dressings. Non-dyadic $k = 3$ is excluded by Banderier–Drmota for $\mathbb{N}$-algebraic DSEs but permitted by signed/annihilation dynamics, which both conserved classes plausibly exhibit. **(ii)** We predict the Manna–C-DP τ-gap from the rank-3 bridge function without fitting parameters and compare to published simulations. The paper also documents three directions — LERW in $d = 3$, TAP complexity in spin glasses, KPZ upper-tail universality — where we *believe* the current CFAC machinery does not reach, with stated structural reasons. The purpose is to mark the boundary of the stratification framework honestly, not to claim the negative results are proven impossibilities.

Contents

1 Introduction

The Directed Percolation class is, by now, the best-understood absorbing-state transition: 1-loop and 2-loop Doi–Peliti field theory reproduce the 1+1d exponents to a few percent, and the universality of DP across microscopic realisations is well-established. Outside DP, the picture is more fragmented. Ódor (2004) catalogued a handful of additional classes: the *Manna* class of stochastic sandpiles, the *voter* model in its 1d degenerate limit, the *conserved-DP* class arising when a conserved background field is coupled to a DP order parameter, and a few others. Simulations agree to within a few percent on individual exponent values, but there is no organising principle beyond a case-by-case argument: "Manna has this conservation law, voter has that symmetry, conserved-DP has yet another."

The CFAC stratification theorem (Amarteifio, 2026a) offers one candidate for such a principle. Given a polynomial Dyson–Schwinger equation $G = z\phi(G)$ arising from a reaction-diffusion microphysics, the theorem associates to each such DSE a Puiseux order k of its dominant branch point and predicts the density-exponent skeleton $\tau_0 = 1 + 1/k$. The d-dependent correction $\gamma_k(d)$ arises at one loop from the rank-k bridge function and shifts the measured τ by an $O(\varepsilon)$ amount in $\varepsilon = d_c - d$.

In this note we test the hypothesis that Manna and conserved-DP share a common $k = 3$ stratum, with the 4% gap between their simulation values $\tau_{\text{Manna}} \approx 1.29$ and $\tau_{\text{C-DP}} \approx 1.35$ accounted for entirely by the difference in their one-loop $\gamma_3(d = 1 + 1)$. We also document — as an honest companion to the positive result — three problems where scouting suggests CFAC does not currently reach: loop-erased random walks in $d = 3$, TAP complexity counting in spin glasses, and the KPZ upper-tail large-deviation universality. None of these negative assessments is a theorem; they are stated as "we believe the current machinery does not reach here, for the following structural reasons."

Contents. Section 2 recalls the stratification theorem. Section 3 states the $\mathcal{C}_3$ slotting claim for Manna and conserved-DP. Section 4 computes the rank-3 bridge $\gamma_3(d = 1 + 1)$ and predicts the τ-gap. Section 5 compares to published simulations. Section 6 documents what currently resists slotting and why. Section 7 concludes.

2 The CFAC stratification theorem

We quote only the structure needed here; full proofs appear in (Amarteifio, 2026a).

Definition 1 (Polynomial DSE)**.** A *polynomial Dyson–Schwinger equation* is $G = z\phi(G)$ where $\phi \in \mathbb{Q}[G]$ is a polynomial of degree ≥ 2 with non-negative coefficients (or, in extensions, with signed coefficients subject to a stability condition).

Theorem 1 (Stratification, (Amarteifio, 2026a))**.** *Let $G(z)$ satisfy $G = z\phi(G)$ with ϕ polynomial. Then the dominant singularity $z_\star$ is a Puiseux branch point of order $k \in \{2, 3, 4, \dots\}$, and the density exponent satisfies*

$$\tau(d) = 1 + \frac{1}{k} + \gamma_k(d), \qquad \gamma_k(d) = c_k\, \varepsilon + O(\varepsilon^2), \quad (1)$$

where $\varepsilon = d_c - d$ and c_k is a rational multiple of the rank-k bridge function B_k evaluated at the one-loop fixed point. ***Banderier–Drmota constraint:*** *for $\mathbb{N}$-algebraic DSEs (non-negative polynomial coefficients), the dominant k lies in the dyadic ladder $\{2, 4, 8, \dots\}$. For signed/annihilation DSEs, non-dyadic integer k is accessible.*

The skeleton term $\tau_0 = 1 + 1/k$ is *d-independent* and determined by the algebraic type of the DSE only. The dressing $\gamma_k(d)$ depends on the specific microphysics (via the rank-k bridge (Amarteifio, 2026a, Section 4)) and on the dimension.

Physical content. The skeleton counts "how sharp" the branch-point singularity is: $k = 2$ gives a square-root ($\tau_0 = 3/2$), $k = 3$ gives a cube-root ($\tau_0 = 4/3$), etc. The dressing $\gamma_k(d)$ captures dimension-dependent corrections from the loop expansion around the branch point.

3 The $\mathcal{C}_3$ slotting claim

Claim (slotting). The Manna and conserved-DP universality classes both sit on the $\mathcal{C}_3$ stratum of Theorem 1: $k = 3$, skeleton $\tau_0 = 4/3$, with different $\gamma_3(d = 1+1)$ dressings accounting for the observed gap between their simulation τ values.

Raw slotting. Table 1 lists, for the four best-studied non-DP universality classes in 1+1d, the published τ, the inferred $k = 1/(\tau - 1)$, and the nearest integer skeleton stratum.

Class	τ	k_{raw}	$\mathcal{C}_k$	$\tau_0 = 1 + 1/k$
DP 1+1d	1.108	9.26	$\mathcal{C}_2$	1.500
Manna 1+1d	1.29	3.45	$\mathcal{C}_3$	1.333
C-DP 1+1d	1.35	2.86	$\mathcal{C}_3$	1.333
voter 1d	$\to 1$	$\to \infty$	$\mathcal{C}_\infty$	1.000

Table 1: Raw slotting of published τ values onto integer strata. Values from (Ódor, 2004; Henkel, Hinrichsen and Lübeck). Manna and C-DP both sit within 3% of the $\mathcal{C}_3$ skeleton.

Signal: Manna and C-DP flank $\mathcal{C}_3$ from opposite sides. The Manna residual $\gamma_3^{\text{Manna}} = \tau_{\text{Manna}} - \tau_0 \approx -0.04$ and the C-DP residual $\gamma_3^{\text{C-DP}} \approx +0.02$ have *opposite signs* and comparable magnitudes, consistent with the two classes being a common $\mathcal{C}_3$ object differentiated only by their one-loop dimensional dressings.

DP sits on $\mathcal{C}_2$ with a full 1-loop dressing. The raw $k_{\text{raw}} = 9.26$ inferred from $\tau_{\text{DP}} = 1.108$ is misleading: DP is an $\mathbb{N}$-algebraic DSE with dominant Puiseux order $k = 2$ (square-root branch), and the 40% gap from $\tau_0 = 3/2$ is the full 1-loop $\gamma_2(d = 1)$ at $\varepsilon = 3$. This is expected, not a failure of the slotting.

Dyadic / non-dyadic distinction. Banderier–Drmota excludes $k = 3$ for $\mathbb{N}$-algebraic DSEs. Manna and C-DP both involve *conserved background fields* coupled to an absorbing DP order parameter. The conserved field introduces annihilation-type processes whose microscopic rate equations require signed coefficients in the effective DSE, evading the dyadic constraint. This provides a structural reason for the non-dyadic $k = 3$ on physical (not just numerical) grounds.

Remark 1. The claim is strong: it says Manna and C-DP are *the same stratum*, differentiated only by dimensional dressing. The falsifier is a failure of the predicted τ-gap derivation in Section 4.

4 The rank-3 bridge and $\gamma_3(d = 1 + 1)$

The dressing $\gamma_k(d)$ in Theorem 1 arises from the rank-k bridge function B_k evaluated at the one-loop DP-like fixed point and integrated over the d-dimensional momentum shell.

Rank-k bridge. From (Amarteifio, 2026a, Prop. 4.3),

$$B_k(d) = \frac{2}{(k-1)!\,(4\pi)^{k/2}}. \quad (2)$$

For $k = 3$, $d = 2$: $B_3(2) = 2/(2! \cdot (4\pi)^{3/2}) = 1/(4\pi)^{3/2} \approx 0.02247$.

One-loop γ_3. [TODO: insert derivation of $\gamma_3(d)$ from the rank-3 bridge and the $\mathcal{C}_3$ one-loop fixed-point amplitude. The result should be of the form $\gamma_3(d) = a_3\varepsilon + O(\varepsilon^2)$ with a_3 a rational multiple of B_3 and an integer combinatorial factor from the cube-root fixed-point profile.]

Prediction. The Manna and conserved-DP τ-gap follows from the difference in their γ_3 evaluations, which in turn depends only on the microscopic rate constants of the two processes. Specifically, Manna's extra reaction $2A \to \emptyset$ (absorbing pair annihilation) and C-DP's conservation law enter through the same one-loop fixed-point amplitude but with different signs. Writing the predicted gap as

$$\tau_{\mathrm{C-DP}} - \tau_{\mathrm{Manna}} = \gamma_3^{\mathrm{C-DP}}(d{=}2) - \gamma_3^{\mathrm{Manna}}(d{=}2), \quad (3)$$

the right-hand side is computable without fitting parameters from Eq. (2) plus the specific one-loop vertex amplitude for each class.

[TODO: numerical prediction, with target simulation value $\Delta\tau_{\mathrm{obs}} = 1.35 - 1.29 = 0.06$.]

5 Cross-checks against simulations

Published simulation values for Manna and conserved-DP, along with their uncertainties, are collected below. We compare to our γ_3-based prediction.

Class	Source	τ	Error bar
Manna	(Ódor, 2004)	1.29	± 0.02
Manna	(Henkel, Hinrichsen and Lübeck)	1.275	± 0.01
C-DP	(Ódor, 2004)	1.35	± 0.03
C-DP	(Lübeck and Hucht, 2002)	1.338	± 0.005

Table 2: Published simulation τ for the two $\mathcal{C}_3$ candidates. The observed gap is $\Delta\tau_{\mathrm{obs}} \approx 0.06 \pm 0.03$.

[TODO: compare prediction from Eq. (3) to the observed gap $\Delta\tau_{\mathrm{obs}}$. Report relative error. The slotting claim is falsifiable: if the prediction is outside the error bars of the simulations, the claim fails.]

Independent check: DP on $\mathcal{C}_2$. The same framework applied to DP with $k = 2$ should give $\tau_{\mathrm{DP}} = 3/2 + \gamma_2(d = 1)$, with $\gamma_2(d = 1) \approx -0.39$. The $\gamma_2(d)$ computation is standard and has been done at 2 loops in the DP literature (Ódor, 2004, Table 4); we use it as an independent check that our γ_k machinery is calibrated correctly.

6 Directions we believe do *not* currently slot

Scouting four further problems from the CFAC candidate list (Amarteifio, 2026b, `docs/problems.md`) returned clear structural reasons why the current stratification framework does not reach them. These are not proven no-gos: each flags a specific extension that would be required. We state them here to mark the boundary of the framework honestly.

6.1 LERW in $d = 3$

Loop-erased random walks in $d = 3$ have a scaling exponent $\nu \approx 1.624$ that has resisted closed form since Kozma (Kozma, 2007) proved it is not equal to 3/2. Wilson's algorithm connects LERW to spanning-tree enumeration, i.e. to the Kirchhoff polynomial that CFAC computes natively. However:

- In the thermodynamic limit the LERW generating function on $\mathbb{Z}^3$ is a lattice Green's function, not algebraic and not D-finite; Theorem 1 does not apply.
- Setting $\tau = \nu$ would require $k = 1/(\nu - 1) \approx 1.60$, neither integer nor dyadic.

We believe LERW 3D is not reachable without extending the stratification framework to non-D-finite generating functions. A legitimate adjacent calculation would be the tropical Kirchhoff decomposition of Wilson-weight expectations on *finite* graphs.

6.2 TAP complexity in spin glasses

The log-count of critical points of a random spin-glass energy landscape — the Bray–Moore / Crisanti–Leuzzi complexity (Bray and Moore, 1980; Crisanti and Leuzzi, 2005), sharpened by Auffinger–Ben Arous–Černý (Auffinger, Ben Arous and Černý, 2013) and Subag (Subag, 2017) for pure p-spin — is controlled by the Kac–Rice formula $\mathbb{E}|\det(H - fI)|$, where H is the Hessian, an $N \times N$ GOE matrix. The CFAC bridge function (Amarteifio, 2026a, Section 4) is scalar: it handles mass-independent scalar self-energies and rank-k polynomial integrands. The object that would describe TAP complexity is a *matrix-valued* bridge — the operator generalisation whose scalar limit recovers Eq. (2) but whose determinantal structure captures the GOE spectral measure. We believe this is a genuine extension requiring new mathematics (free probability, R-transforms), not a mechanical tensor dressing of the existing bridge. TAP complexity is parked pending that extension.

6.3 KPZ upper-tail universality

The KPZ equation in 1+1d has an upper-tail rate function of shape $|h|^{3/2}$ with a coefficient known only for integrable weight distributions (Corwin and Ghosal, 2020; Krajenbrink, 2020). Universality beyond integrable weights is one of the top open problems in integrable probability.

- Structural match: $\tau = 3/2$ is exactly the $\mathcal{C}_2$ skeleton (square-root branch), the generic Puiseux class of a polynomial DSE. The tilted-DSE machinery in `tilted.py` handles SCGFs of polynomial DSEs. Reproducing the *exponent* is, structurally, cheap.

- Missing machinery: the *coefficient* requires a bridge function for disorder-averaged random-weight transfer matrices. No such bridge is currently implemented; the existing bridges cover deterministic Doi–Peliti + MSR only.

We believe KPZ upper-tail universality is a "cautious go" for the scoping layer (show that $\tau = 3/2$ comes out of the tilted directed-polymer DSE at the $\mathcal{C}_2$ stratum) and a research programme for the full universality proof.

6.4 Why listing these matters

The stratification theorem, as stated, is narrow: it applies to polynomial DSEs whose generating functions are algebraic or D-finite. Each negative case above identifies a specific structural feature (non-D-finite GF, matrix-valued bridge, disorder-averaged bridge) that the current framework does not handle. We list them here rather than in a separate note because the positive result (Manna/C-DP on $\mathcal{C}_3$) is only interpretable against the clear boundary of what the framework currently covers.

7 Discussion

If the γ_3 computation in Section 4 reproduces the Manna–C-DP τ-gap within simulation error bars, the stratification theorem provides a genuinely new organising principle for non-DP universality: classes are grouped by their skeleton k, with observable τ differences arising from calculable dressings rather than being independent phenomenological parameters.

What is *not* claimed. We do not claim to prove the 1+1d Manna or C-DP exponents analytically. We claim that the *slotting* — the organisation of the universality landscape by $\mathcal{C}_k$ — is a correct coarse structure, falsifiable by the γ_3 prediction. A failure of that prediction would rule out the slotting.

Next steps. (i) Complete the $\gamma_3(d = 1+1)$ calculation in Section 4 and confirm or falsify the τ-gap prediction. (ii) Extend to the voter class at $k \to \infty$ with log corrections. (iii) Test the slotting on a further non-DP class not used in the hypothesis formation (parity-conserving branching annihilation random walks, for instance).

References

Amarteifio, S., 2026. *CFAC theorem: stratification of Dyson–Schwinger equations by Puiseux order.* Percolation Labs preprint.

Amarteifio, S., 2026. Candidate problems for the CFAC framework. `rdft/docs/problems.md`.

Auffinger, A., Ben Arous, G., and Černý, J., 2013. Random matrices and complexity of spin glasses. *Comm. Pure Appl. Math.* **66**, 165.

Bray, A. J. and Moore, M. A., 1980. Metastable states in spin glasses. *J. Phys. C* **13**, L469.

Corwin, I. and Ghosal, P., 2020. Lower tail of the KPZ equation. *Duke Math. J.* **169**, 1329.

Crisanti, A. and Leuzzi, L., 2005. Spherical 2+p spin-glass model. *Phys. Rev. B* **73**, 014412.

Henkel, M., Hinrichsen, H. and Lübeck, S., 2008. *Non-Equilibrium Phase Transitions, Vol. 1: Absorbing Phase Transitions.* Springer.

Krajenbrink, A., 2020. From Painlevé transcendents to the KPZ equation. Ph.D. thesis, Ecole Normale Supérieure, Paris.

Kozma, G., 2007. The scaling limit of loop-erased random walk in three dimensions. *Acta Math.* **199**, 29.

Lübeck, S. and Hucht, A., 2002. Absorbing state phase transition in conserved DP. *J. Phys. A* **35**, 4853.

Ódor, G., 2004. Universality classes in non-equilibrium lattice systems. *Rev. Mod. Phys.* **76**, 663.

Subag, E., 2017. The complexity of spherical p-spin models. *Ann. Probab.* **45**, 3385.